

Prompt used:

Generate C code to implement the following requirements:

Objective

Write a C program that picks a secret number between 1 and 10.

The user gets up to 3 tries to guess the number.

Requirements

- Use a fixed secret number (for example, 7).
- Use printf to prompt the user and scanf to read their guess.
- Use a while or for loop to let the user guess up to 3 times.
- Use if-else statements to check if the guess is too high, too low, or correct.
- Stop the loop immediately if the user guesses the correct number.
- Print a final win or lose message.

Example Output

text

Guess a number between 1 and 10.

Attempt 1/3. Enter your guess: 5

Too low! Try again.

Attempt 2/3. Enter your guess: 8

Too high! Try again.

Attempt 3/3. Enter your guess: 7

Correct! You win!

ChatGPT:

```
#include <stdio.h>
```

```
int main(void) {
    int secretNumber = 7;
    int guess;
    int attempts = 0;
    int won = 0;

    printf("Guess a number between 1 and 10.\n");

    while (attempts < 3) {
        attempts++;

        printf("Attempt %d/3. Enter your guess: ", attempts);
        scanf("%d", &guess);

        if (guess == secretNumber) {
            printf("Correct! You win!\n");
            won = 1;
            break;
        } else if (guess < secretNumber) {
            printf("Too low! Try again.\n");
        } else {
            printf("Too high! Try again.\n");
        }
    }

    if (!won) {
        printf("You lose! The secret number was %d.\n", secretNumber);
    }

    return 0;
}
```

Gemini:

```
#include <stdio.h>
```

```
int main() {
    int secret = 7;
    int guess;
    int max_tries = 3;
    int has_won = 0;

    printf("Guess a number between 1 and 10.\n");

    for (int i = 1; i <= max_tries; i++) {
        printf("Attempt %d/%d. Enter your guess: ", i, max_tries);
        scanf("%d", &guess);

        if (guess == secret) {
            printf("Correct! You win!\n");
            has_won = 1;
            break;
        } else if (guess < secret) {
            if (i < max_tries) {
                printf("Too low! Try again.\n");
            }
        } else {
            if (i < max_tries) {
                printf("Too high! Try again.\n");
            }
        }
    }

    if (!has_won) {
        printf("Game over! You've used all %d attempts. The secret
number was %d.\n", max_tries, secret);
    }

    return 0;
}
```

}

ChatGPT Analysis

- a. ChatGPT
- b. chatgpt.com
- c. See above
- d. See above
- e. Further analysis
 - i. Correctness - Does not handle string inputs
 - ii. Time complexity - While loops are less efficient than for loops
 - iii. Space complexity - Could use macros for non changing variables
 - iv. Maintainability - There are no comments

Gemini Analysis

- a. Gemini
- b. gemini.google.com
- c. See above
- d. See above
- e. Further analysis
 - i. Correctness - Does not handle string inputs
 - ii. Time complexity - For loops are the more efficient solution when compared to while loops
 - iii. Space complexity - Could use macros for non changing variables
 - iv. Maintainability - There are no comments

I will use the Gemini code, as for loops are more efficient than while loops. I will add handling of bad inputs, implement macros for constants, and add comments.